

Lección 22

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Plan de trabajo

► Formulación hamiltoniana

- Ecuaciones de Hamilton de la partícula relativista
- Teoría de perturbaciones

► Nuestro Syllabus:

FIS-310:

UNIDAD 5. Formulación Hamiltoniana

- 5.1. Transformaciones de Legendre
- 5.2. Ecuaciones de Hamilton
- 5.3. Ecuaciones de Hamilton de la Partícula Relativista
- 5.4. Corchete de Poisson
- 5.5. Transformaciones Canónicas
- 5.6. Ecuaciones de Hamilton-Jacobi
- 5.7. Teoría de perturbaciones

FIS-345:

Unidad IV: Formulación Hamiltoniana General

- Transformaciones canónicas
- Variables de ángulo y acción. Invariantes Adiabáticos

Unidad V: Teoría de perturbaciones y pequeñas oscilaciones

- Teoría canónica de perturbaciones
- Caos en sistemas Hamiltonianos

FIS-426:

4) Formulación Hamiltoniana General

- Paréntesis de Poisson
- Transformaciones canónicas
- Variables de ángulo y acción
- Teoría de perturbaciones
- Caos en sistemas Hamiltonianos

- **FIS-345:** Tema pendiente: invariantes adiabáticos. Vamos a verlos después de teoría de perturbaciones (son cantidades que se conservan aproximadamente)

Relativistic particle in Hamilton-Jacobi formalism

Now we are going to extend the canonical formalism for relativistic system

—Is straightforward, but there are some points which require attention

Bibliography: See Sec 8.4 in Goldstein and Sec 12 in:

O.D. Johns, “*Analytical Mechanics for Relativity and Quantum mechanics*”, Oxford University Press (2005),

ISBN: 019856726X, 9780198567264 (Hbk)

Construct the Hamiltonian and apply the Hamilton-Jacobi formalism for relativistic particle with momentum m moving in the external electric field (potential $\phi(\mathbf{x})$). Find the most general potential which allows separation of variables in Cartesian coordinates

Relativistic particle in Hamilton-Jacobi formalism

► Lagrangian:

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q\phi(\mathbf{x})$$

▷ Generalized momentum & Hamiltonian:

$$\mathbf{p} = \frac{m\mathbf{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$H = \mathbf{p} \cdot \mathbf{v} - L = \sqrt{p^2 c^2 + m^2 c^4} + q\phi(\mathbf{x})$$

Hamilton-Jacobi equation: $E \rightarrow \frac{\partial S}{\partial t}$, $\mathbf{p} \rightarrow \nabla S$

$$(\nabla S)^2 c^2 + m^2 c^4 = (E - q\phi(\mathbf{x}))^2 = \left(\frac{\partial S}{\partial t} - q\phi(\mathbf{x}) \right)^2$$

Relativistic particle in Hamilton-Jacobi formalism

- Assume the external potential $\phi \neq 0$
- Separation of variables in Cartesian coordinates:

$$S = -Et + f_1(x_1) + f_2(x_2) + f_3(x_3)$$

$$\underbrace{(f_1'(x_1))^2}_{=\text{const}=p_1^2} + \underbrace{(f_2'(x_2))^2}_{=\text{const}=p_2^2} + \underbrace{(f_3'(x_3))^2}_{=\text{const}=p_3^2} + m^2 c^2 = \frac{(E - q\phi(\mathbf{x}))^2}{c^2}$$

The separation of variables will work only if $E - q\phi(\mathbf{x})$ depends on one of the variables x_1, x_2 or x_3

- Assume $\phi = \phi(x_1)$ for definiteness

$$\Rightarrow S = -Et + \sum_{i \neq 1} p_i x_i + \int^{x_1} dx_1 \sqrt{\frac{(E - q\phi(x_1))^2}{c^2} - p_2^2 - p_3^2}$$

$$p_2 = \text{const}, \quad p_3 = \text{const}$$

Relativistic particle in canonical formalism

Analyze if Lorentz transformation may be considered as a canonical transformation from the point of view of Hamiltonian formalism. If yes, find the generating function of this transformation.

Help: You may consider only the Lorentz boost in x -direction given by

$$x' = \frac{x - V t}{\sqrt{1 - V^2/c^2}},$$

$$ct' = \frac{ct - x V/c}{\sqrt{1 - V^2/c^2}},$$

$$y' = y, \quad z' = z.$$

Please let me know your ideas ...

Relativistic particle in canonical formalism

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}, \quad x^{\mu} = \Lambda^{\mu}_{\nu} x'^{\nu}, \quad \Lambda^{\mu}_{\nu} \Lambda^{\nu}_{\beta} = \delta^{\mu}_{\beta}$$

Help: You may consider the Lorentz boost in x-direction given by

$$x' = \gamma(x - \beta ct), \quad ct' = \gamma(ct - \beta x), \quad y' = y, \quad z' = z, \quad \beta = \frac{v}{c}, \quad \gamma = 1 / \sqrt{1 - \beta^2}$$

$$p' = \gamma(p_x - \beta E), \quad E' = \gamma(E - \beta p_x), \quad p'_y = p_y, \quad p'_z = p_z,$$

Naive approach:

– Calculate pairwise Poisson brackets

$$[p_x, q_x] = \gamma^2 \neq 1,$$

for others we get as required

$$[p_a, q_b] = \delta_{ab}, \quad [p_a, p_b] = 0, \quad [q_a, q_b] = 0$$

Relativistic particle in canonical formalism

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}, \quad x^{\mu} = \Lambda^{\mu}_{\nu} x'^{\nu}, \quad \Lambda^{\mu}_{\nu} \Lambda^{\nu}_{\beta} = \delta^{\mu}_{\beta}$$

Help: You may consider the Lorentz boost in x -direction given by

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– We'll ignore variables y, z for simplicity

Another (Naive) attempt:

Lorentz transformation is time-dependent *point* transformation

– is canonical if momenta transform as ($q \equiv x, \quad Q \equiv x'$)

$$\left(\frac{\partial P_b}{\partial p_a} \right)_{q,p} = \left(\frac{\partial q_a}{\partial Q_b} \right)_{Q,P} = \Lambda^a_b, \quad \begin{pmatrix} \gamma\beta & & \\ & 1 & \\ & & 1 \end{pmatrix}_{a,b}$$

Caution:

– Lorentz transformation mixes coordinates and time

– time is not treated as coordinate in canonical formalism!

... we always assumed that $t' = t$ in Galileo's approach

– Know that energy (H) and momenta form components of 4-vector

Relativistic particle in canonical formalism

- Start with minimal action principle in different inertial frames, related by Lorentz boost in direction of axis $\hat{x}$ ("axis 1")

$$\delta S = \delta \int dt \left(\sum_a p_a \dot{q}_a - H(p, q, t) \right) = \delta \int dt' \left(\sum_a P_a \dot{Q}_a - \tilde{H}(P, Q, t) \right) = \min$$

$$\Rightarrow \left(\sum_a p_a dq_a - H(p, q, t) dt \right) = \left(\sum_a P_a dQ_a - \tilde{H}(P, Q, t) dt' \right) + dF(q, Q, t)$$

—In general $t = t(t', Q_a)$ and $t' = t'(t, q_a)$. so

$$dt' = \frac{\partial t'}{\partial t} dt + \frac{\partial t'}{\partial q_a} dq_a, \quad dt = \frac{\partial t}{\partial t'} dt' + \frac{\partial t}{\partial Q_a} dQ_a$$

$$\Rightarrow p_a = \frac{\partial F}{\partial q_a}, \quad P_a = -\frac{\partial F}{\partial Q_a} - \frac{\partial F}{\partial t} \frac{\partial t}{\partial Q_a}, \quad \tilde{H} = \frac{dt}{dt'} H + \frac{dt}{dt'} \frac{\partial F}{\partial t}$$

If we assumed that $F = F(q, Q, t')$, we would get

$$\Rightarrow p_a = \frac{\partial F}{\partial q_a} + \frac{\partial F}{\partial t'} \frac{\partial t'}{\partial q_a}, \quad P_a = -\frac{\partial F}{\partial Q_a}, \quad \tilde{H} = \frac{dt}{dt'} H + \frac{\partial F}{\partial t'}$$

Relativistic particle in canonical formalism

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Recall Lorentz transformation of energy-momentum:

$$E' = \gamma E - \gamma V P_1, \quad P'_1 = \gamma P_1 - \gamma V E, \quad P'_{2,3} = P_{2,3}$$

—Problem is far more complicated than in case of Galileo's transformation, need to reformulate everything in covariant form

Relativistic particle in canonical formalism

Goldstein, Chapter 8

- Lecture 12: explicit integration over time in $S = \int dt L$ breaks “equivalence” of time and space coordinates.
 - ▶ May define covariant lagrangian Λ , such that
$$S = \int d\tau \Lambda, \quad \Lambda = \dot{x}_0 L(\mathbf{x}, \dot{\mathbf{x}}/\dot{x}_0), \quad \dot{q} \equiv dq/d\tau$$
- There is similar “extended” Hamiltonian formulation which tries to keep Lorentz covariance
 - ▶ Consider x_0 as 4th coordinate and p_0 as conjugate momentum:

$$p_0 = \frac{\partial \Lambda}{\partial \dot{x}_0} = L + \dot{x}_0 \frac{\partial L(\mathbf{x}, \dot{\mathbf{x}}/\dot{x}_0)}{\partial \dot{x}_0}$$

$$\mathcal{H} = p^\mu \dot{x}_\mu - \Lambda = p^\mu \dot{x}_\mu - \Lambda$$

where $\mathcal{H} \equiv \gamma(H + p_0)$ is extended Hamiltonian of the system

- ▶ Canonical equations in Lorentz covariant form:

$$\dot{p}_\mu \equiv [\mathcal{H}, p_\mu] = -\frac{\partial \mathcal{H}}{\partial q_\mu}, \quad \dot{q}_\mu \equiv [\mathcal{H}, q_\mu] = \frac{\partial \mathcal{H}}{\partial p_\mu}, \quad \mu = 0, 1, 2, 3$$

- ▶ We can also get the same result starting from
$$S = \int d\tau \Lambda(\mathbf{x}^\mu, d\mathbf{x}^\mu/d\tau, \tau)$$
and repeating derivation of the canonical equations from Lecture 14

Relativistic particle in canonical formalism

$$\dot{p}_\mu \equiv [\mathcal{H}, p_\mu] = -\frac{\partial \mathcal{H}}{\partial q_\mu}, \quad \dot{q}_\mu \equiv [\mathcal{H}, q_\mu] = \frac{\partial \mathcal{H}}{\partial p_\mu}, \quad \mu = 0, 1, 2, 3$$

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Relativistic particle in canonical formalism

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where $\mathcal{H} \equiv H(\mathbf{p}, \mathbf{q}, t) + p_0$ is extended Hamiltonian of the system

- So far we took derivative with respect to lab frame
 - ▶ Sometimes evolution w.r.t. proper time τ is used
 - ▶ Since $d/d\tau = \gamma d/dt$, should redefine the hamiltonian as $\tilde{\mathcal{H}} \equiv \gamma(H + p_0)$
 - We can also get the same result starting from $S = \int d\tau \Lambda(x^\mu, dx^\mu/d\tau, \tau)$ and repeating derivation of the canonical equations from Lecture 14
- ▶ For more details see e.g. Sec 12 in:

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Perturbation theory.

- Up to now we considered that we *a/ways* can solve the EOMs exactly (or at least get formal/numerical solutions)

* Not always is possible, in some problems none of the methods works

- In some problems the hamiltonian H is close to analytically solvable H_0
... may expect that deviations

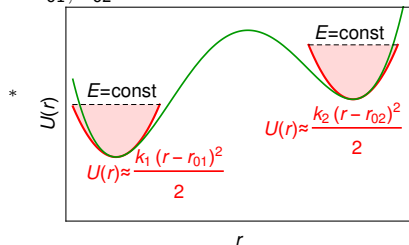
$$\delta \mathbf{r} = \mathbf{r}_H(t) - \mathbf{r}_{H_0}(t),$$

should be small for any t , if they are close enough at some initial moment t_0
[here $\mathbf{r}_H(t)$, $\mathbf{r}_{H_0}(t)$ are trajectories found with hamiltonians H or H_0]

Example - harmonic oscillator

► Simplest case:

Potential with local minima at points r_{01}, r_{02}



Near local minimum, can approximate the hamiltonian $H = \frac{p^2}{2m} + U$ with hamiltonian of harmonic oscillator,

$$H_0 \approx \frac{p^2}{2m} + \frac{k}{2} (r - r_0)^2$$

or in N dimensions ($\sum_{ij}$)

$$H_0 \approx \frac{p_i (\hat{m}^{-1})_{ij} p_j}{2} + \frac{k_{ij}}{2} (r - r_0)_i (r - r_0)_j$$

* Formal solutions for $\mathbf{r}_{H_0}(t)$: harmonic oscillations near the equilibrium point r_0
Can treat the difference $\delta H = H - H_0$ as a perturbation ("anharmonic terms" $\sim (x - x_0)^{3,4}$)

⇒ In full theory the trajectory $\mathbf{r}_H(t)$ should be close to $\mathbf{r}_{H_0}(t)$

– the method does NOT work near local maximum
(or saddle-point in N dimensions)

Extension to other “solvable” potentials

- Assume the potential $U(\mathbf{r})$ might be decomposed as $U(\mathbf{r}) = U_0(\mathbf{r}) + \Delta U(\mathbf{r})$ where U_0 is analytically solvable potential, and $\Delta U \ll U$
- Equation of motion (2nd law of Newton/Euler-Lagrange)

$$m\ddot{\mathbf{r}} = -\nabla U(\mathbf{r}) \quad (1)$$

- We may try to solve the equations making perturbative expansion,

$$\mathbf{r}(t) \rightarrow \mathbf{r}_0(t) + \delta\mathbf{r}(t) \quad \text{where} \quad m\ddot{\mathbf{r}}_0 = -\nabla U_0(\mathbf{r}_0(t))$$

$$\Rightarrow m\delta\ddot{r}_i = - \left. \frac{\partial \Delta U}{\partial r_i} \right|_{\mathbf{r} \equiv \mathbf{r}_0(t)} - \left. \frac{\partial^2 U_0}{\partial r_k \partial r_i} \right|_{\mathbf{r} = \mathbf{r}_0(t)} \delta r_k$$

—usually (2) is time-dependent equation for $\mathbf{r}_1(t)$, only numerical solutions

$$\text{Eq. (2)} \quad \Rightarrow \quad m\delta\ddot{r}_i + k_{ij}(t)\delta r_j = - \left. \frac{\partial \Delta U}{\partial r_i} \right|_{\mathbf{r} \equiv \mathbf{r}_0(t)} . \quad k_{ij}(t) = \left. \frac{\partial^2 U_0}{\partial r_i \partial r_j} \right|_{\mathbf{r} = \mathbf{r}_0(t)}$$

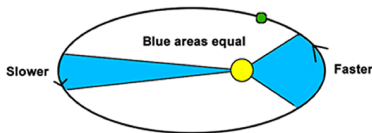
—resembles equation for forced oscillations of harmonic oscillator, but with time-dependent $k_{ij}(t)$

—expect perturbation theory will not work when particle passes the region where k_{ij} is

Challenges of “direct” approach

Motion in central potential, Lect. 7:

$$\int^r \frac{dr}{\sqrt{\frac{2}{m} (E - U_{\text{eff}}(r))}} = t - t_0$$



$$\phi - \phi_0 = \int^r dr \frac{M}{m r^2 \sqrt{\frac{2}{m} (E - U_{\text{eff}}(r))}}$$

$$U_{\text{eff}} = U(r) + \frac{M^2}{2mr^2}$$

► If $U = U_0(r) + \Delta U(r)$, perturbative expansion over ΔU yields

$$\int \frac{dr}{\sqrt{(E - U_{0,\text{eff}}(r))}} - \int \frac{dr}{(E - U_{0,\text{eff}}(r))^{3/2}} \Delta U(r) + \dots$$

get **unintegrable singularities** near the points where $E = U_{0,\text{eff}}(r)$ (when the particle passes near the turning point).



⇒ (Direct) Perturbation theory fails/becomes inapplicable

Canonical perturbation theory (idea)

Main ideas:

- We do NOT assume that trajectories are close,

$$\delta \mathbf{r}(t) \equiv \mathbf{r}(t) - \mathbf{r}_0(t)$$

are small, nor make expansion over $\delta \mathbf{r}$

- Assume that we know exact solution of equations of motion in $U_0(\mathbf{r})$.

- This solution in general depends on some constants (integrals of motion) α_a

- determine shape, orientation of trajectory
- should be fixed from initial conditions

- We assume that due to perturbation, these constants α_a become slowly-changing functions of time, $\alpha_a = \alpha_a(t)$, and fix this dependence

Canonical perturbation theory

Hamilton-Jacobi equation

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a}, q_a, t\right) = 0 \quad (1)$$

- Expect that $S = f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t) + A$ where $\alpha_i, A = \text{const}$
- Case of conservative systems: $S = f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N) - E t$
- Let's consider $f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$ as a generating function $F(q, P, t)$ which relates old and new coordinates

$$p_a = \frac{\partial f}{\partial q_a}, \quad Q_a \equiv \beta_a = \frac{\partial f}{\partial \alpha_a}$$

$$\tilde{H} = H + \frac{\partial f}{\partial t} = 0 \quad (!!!)$$

$$\Rightarrow \dot{\alpha}_a = 0, \quad \dot{\beta}_a = 0,$$

$$\Rightarrow \alpha_a = \text{const}, \quad \beta_a = \text{const},$$

Canonical perturbation theory

- Assume $H = H_0 + \Delta H$, where H_0 is analytically solvable and $\Delta H \ll H_0$
- Assume we know $f_0(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$ of solvable problem and use it as generating function

$$H = H_0 + \Delta H$$

$$\Rightarrow \tilde{H} = H + \frac{\partial f_0}{\partial t} = \Delta H \neq 0 (!!!)$$

$\Rightarrow$ Canonical equations for (α, β) :

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta}, \quad \dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} \quad (2)$$

▷ Usually we can't solve (2) exactly

... but we know that ΔH is small, so can assume that $\alpha(t), \beta(t)$ *change slowly*:

$$\alpha \approx \alpha_0 + \alpha_1(t), \quad |\alpha_1| \ll |\alpha_0|$$

$$\beta \approx \beta_0 + \beta_1(t), \quad |\beta_1| \ll |\beta_0|$$

α_0, β_0 are found from solution of HJ with hamiltonian H_0

Canonical perturbation theory (algorithm-summary)

Assume $H = H_0 + \Delta H$, where H_0 is analytically solvable and $\Delta H \ll H_0$

- 1 Solve Hamiltonian or Hamilton-Jacobi equation for H_0 , get trajectory

$$\eta(t, \alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N) = \{ \mathbf{q}_i^{(0)}(t, \alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N), \mathbf{p}_i^{(0)}(t, \alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N) \}$$

- 1 have $2N$ constants since have $2N$ 1st order canonical equations (or N second-order Lagrangian equations)
- 2 Treat $\alpha_1, \dots, \alpha_N$ as “canonical momenta” - these variables appear explicitly in separation of variables in Hamilton-Jacobi, determine the shape of trajectory, its size (energy E , angular momentum M , ...)
- 3 variables $\beta_1, \dots, \beta_N$ are canonically conjugate “coordinates” if we consider as generating function

$$S_0(q_1, \dots, q_N, \alpha_1, \dots, \alpha_N) = \int^q dt \left[\mathbf{p}^{(0)} \dot{\mathbf{q}}^{(0)} - H_0(t, \alpha_1, \dots, \alpha_N) \right], \quad \beta_a = \frac{\partial S_0}{\partial \alpha_a}$$

... definition looks sophisticated, but in reality it isn't:

- 1 E.g., in central potential trajectories depend on $t - t_0$, $\varphi - \varphi_0$, etc. The parameters $\beta_1, \dots, \beta_N$ correspond to parameters φ_0 , t_0 , ... which determine orientation of trajectory in space, time, ...

Canonical perturbation theory (algorithm-summary)

- 1 Variables $\alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N$ are constants in system with hamiltonian H_0 , but NOT constants in theory with hamiltonian H
- 2 Substitute the trajectory into expression for $\Delta H(t, \mathbf{q}, \mathbf{p})$ (which now becomes function of time t and parameters $\alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N$)
- 3 Solve canonical equations for (α, β) :

$$\dot{\alpha}_a = -\frac{\partial \Delta H}{\partial \beta_a}, \quad \dot{\beta}_a = \frac{\partial \Delta H}{\partial \alpha_a},$$

- 1 Usually we can't solve (2) exactly
- 2 ... but we know that ΔH is small, so can assume that $\alpha(t), \beta(t)$ *change slowly*:

$$\alpha \approx \alpha_0 + \alpha_1(t) + \alpha_2(t) + \dots, \quad |\alpha_2| \ll |\alpha_1| \ll |\alpha_0|$$

$$\beta \approx \beta_0 + \beta_1(t) + \beta_2(t) + \dots, \quad |\beta_2| \ll |\beta_1| \ll |\beta_0|$$

α_0, β_0 are constants, and $\alpha_1, \dots, \beta_1, \dots$ can be found using perturbative expansion over ΔH .

Canonical equations - symplectic notations

- Have seen that can rewrite canonical equations in terms of “joined” vector η in $2N$ phase space

$$\eta = \begin{pmatrix} q_1 \\ \dots \\ q_N \\ p_1 \\ \dots \\ p_N \end{pmatrix}, \quad \dot{\eta} = [H, \eta] = J \frac{\partial H}{\partial \eta},$$

- $\partial H / \partial \eta \rightarrow \vec{\nabla} H$ in $2N$ space

$$J = \begin{pmatrix} & I_N \\ -I_N & \end{pmatrix}$$

$$[\eta, \eta] = J \quad [\eta_a, \eta_b] = J_{ab}$$

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta}, \quad \dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} \quad (2)$$

Can rewrite (2) as

$$\dot{\gamma} = [\Delta H, \gamma] = J \frac{\partial \Delta H}{\partial \gamma},$$

where

$$\gamma = \begin{pmatrix} \beta_1 \\ \dots \\ \beta_N \\ \alpha_1 \\ \dots \\ \alpha_N \end{pmatrix},$$

Can solve iteratively $\gamma(t) = \gamma_0 + \gamma_1(t) + \dots$

Example

A particle of mass m moves in the harmonic oscillator potential $U = m\omega^2 x^2/2$. Apply the perturbation theory developed in the previous slides treating potential energy U as perturbation.

Example

- Free hamiltonian:

$$H_0 = \frac{p^2}{2m}$$

$$S_0 = -\frac{\alpha^2}{2m}t + \underbrace{\alpha x}_{f_0(x, \alpha)}$$

$$Q_0 \equiv \beta = \frac{\partial S}{\partial \alpha} = x - \frac{\alpha}{m}t$$

- Introduce perturbation:

$$\Delta H = \frac{m\omega^2 x^2}{2} = \frac{m\omega^2}{2} \left(\beta + \frac{\alpha}{m}t \right)^2$$

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{m\omega^2}{2} \left(\beta + \frac{\alpha}{m}t \right) \quad (1)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{\omega^2}{2}t \left(\beta + \frac{\alpha}{m}t \right) = -\frac{t}{m}\dot{\alpha} \quad (2)$$

- We will solve (1,2) both exactly & iteratively (perturbatively):

Example

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{m\omega^2}{2} \left(\beta + \frac{\alpha}{m} t \right) \quad (1)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{\omega^2}{2} t \left(\beta + \frac{\alpha}{m} t \right) = -\frac{t}{m} \dot{\alpha} \quad (2)$$

Exact solution: from (1)

$$\ddot{\alpha} = -\frac{m\omega^2}{2} \left(\underbrace{\dot{\beta} + \frac{\dot{\alpha}}{m} t}_{=0 \text{ from (2)}} + \frac{\alpha}{m} \right) = -\omega^2 \alpha$$

$$\Rightarrow \alpha(t) = \alpha_0 \cos(\omega t + \phi_0)$$

$$(1) \Rightarrow \beta = -\frac{\alpha}{m} t - \frac{2\dot{\alpha}}{m\omega^2} = -\frac{\alpha_0}{m} t \cos(\omega t + \phi_0) + \frac{2\alpha_0}{m\omega} \sin(\omega t + \phi_0)$$

$$\Rightarrow x(t) = \frac{\alpha(t)}{m} t + \beta(t) = -\frac{2}{m\omega^2} \dot{\alpha} = \frac{2}{m\omega} \alpha_0 \sin(\omega t + \phi_0)$$

-oscillations with frequency ω as expected

Example

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -m\omega^2 \left(\beta + \frac{\alpha}{m} t \right) \quad (1)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \omega^2 t \left(\beta + \frac{\alpha}{m} t \right) = -\frac{t}{m} \dot{\alpha} \quad (2)$$

Iterative (perturbative) solution:

$$\alpha \approx \alpha_0 + \alpha_1(t), \quad |\alpha_1| \ll |\alpha_0|$$

$$\beta \approx \beta_0 + \beta_1(t), \quad |\beta_1| \ll |\beta_0|$$

we'll assume that $\beta_0 = 0$ for simplicity (choice of origin), so from (1,2)

$$\dot{\alpha}_1 = -m\omega^2 \left(\frac{\alpha_0}{m} t \right) \Rightarrow \alpha_1 = - \left(\alpha_0 \omega^2 \frac{t^2}{2} \right)$$

$$\dot{\beta}_1 = \left(\frac{\alpha_0 \omega^2}{m} t^2 \right) \Rightarrow \beta_1 = \left(\frac{\alpha_0 \omega^2}{m} \frac{t^3}{3} \right)$$

$$\Rightarrow \alpha \approx \alpha_0 \left(1 - \frac{\omega^2 t^2}{2} \right), \quad \beta \approx \beta_1(t)$$

$$x = \beta_1 + \frac{\alpha_1}{m} t = \frac{\alpha_0}{m\omega} \left(\omega t - \frac{\omega^3 t^3}{6} \right) \quad p = m\dot{x} \approx \alpha_0 \left(1 - \frac{\omega^2 t^2}{2} \right)$$

Example

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -m\omega^2 \left(\beta + \frac{\alpha}{m}t \right) \quad (1)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \omega^2 t \left(\beta + \frac{\alpha}{m}t \right) = -\frac{t}{m} \dot{\alpha} \quad (2)$$

▷ 2nd order of perturbation theory: replace α, β in right-hand side with α_1, β_1

$$\dot{\alpha}_2 = -\alpha_0 \omega^2 \left(t - \frac{\omega^2 t^3}{6} \right) \quad \Rightarrow \quad \alpha_2 = \alpha_0 - \frac{\omega^2 \alpha_0 t^2}{2} + \frac{\omega^4 \alpha_0 t^4}{24}$$

$$\dot{\beta}_2 = \alpha_0 \frac{\omega^2}{m} \left(t^2 - \frac{\omega^2 t^4}{6} \right) \quad \Rightarrow \quad \beta_2 = \frac{\alpha_0}{m} \frac{\omega^2}{2} \left(\frac{t^3}{3} - \frac{\omega^2 t^5}{30} \right)$$

$$x = \beta_2 + \frac{\alpha_2}{m} t = \frac{\alpha_0}{m\omega} \left(\omega t - \frac{\omega^3 t^3}{6} + \frac{\omega^5 t^5}{120} \right) \approx \frac{\alpha_0}{m\omega} \sin \omega t$$

$$p = m\dot{x} \approx \alpha_0 \left(1 - \frac{\omega^2 t^2}{2} + \frac{\omega^4 t^4}{24} \right) \approx \alpha_0 \cos \omega t$$

⇒ perturbative series reproduces Taylor expansion of trigonometric functions, in agreement with expectations

Example 2

A particle of mass m moves in the anharmonic oscillator potential $U = k q^2/2 + \lambda q^4/4$. Apply the perturbation theory treating the anharmonic term $\sim \mathcal{O}(q^4)$ as perturbation

Result for harmonic term (previous lectures)

Hamilton-Jacobi equation

$$\frac{\partial S_0}{\partial t} + H_0 \left(p_a = \frac{\partial S}{\partial q_a}, q_a, t \right) = 0$$

• $S = f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t) + A$
where α_i , $A = \text{const}$

• Case of conservative systems: $S_0 = f_0(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N) - E t$

$$E = H_0 = -\frac{\partial S_0}{\partial t} \Rightarrow f_0(q) - E t$$

$$H = \frac{p^2}{2m} + \frac{k q^2}{2} = \frac{1}{2m} \left(\frac{df_0}{dq} \right)^2 + \frac{k q^2}{2} = E$$

$$\Rightarrow f_0 = \int dq \sqrt{2m \left(E - \frac{k q^2}{2} \right)} =$$

$$= E \sqrt{\frac{m}{k}} \arctan \left(\frac{q \sqrt{k}}{\sqrt{2E - k q^2}} \right) + \frac{q}{2} \sqrt{m (2E - k q^2)} + S_0$$

• Free constants: $\left(\underbrace{E}_{\alpha}, S_0 \right)$

$$\Rightarrow \frac{\partial S_0}{\partial \alpha} = \underbrace{\sqrt{\frac{m}{k}}}_{1/\omega} \arctan \left(\frac{\sqrt{k q^2}}{\sqrt{2E - k q^2}} \right) - t = 0$$

$$\sqrt{\frac{k q^2}{2E - k q^2}} = \tan(\omega t)$$

$$1 + \tan^2 z = \frac{1}{\cos^2 z} \Rightarrow \cos^2 z = \frac{1}{1 + \tan^2 z}$$

$$\frac{2E - k q^2}{2E} = \cos^2 \omega t \Rightarrow q(t) = \sqrt{\frac{2E}{k}} \sin \omega t$$

Example 2

► Solution of 0-order problem:

$$q_0(t) = \sqrt{\frac{2\alpha}{k}} \sin(\omega t + \beta), \quad \omega \equiv \sqrt{\frac{k}{m}}$$

● Perturbation hamiltonian:

$$\Delta H = \frac{\lambda q_0^4}{4} = \frac{\lambda \alpha^2}{k^2} \sin^4(\omega t + \beta)$$

● Canonical equations in terms of α, β :

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{4\lambda \alpha^2}{k^2} \sin^3(\omega t + \beta) \cos(\omega t + \beta)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{2\lambda \alpha}{k^2} \sin^4(\omega t + \beta)$$

● No known analytical solutions

– Develop perturbation theory

Example 2

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{4\lambda\alpha^2}{k^2} \sin^3(\omega t + \beta) \cos(\omega t + \beta), \quad \dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{2\lambda\alpha}{k^2} \sin^4(\omega t + \beta)$$

–Develop perturbation theory:

$$\alpha \approx \alpha_0 + \alpha_1(t), \quad |\alpha_1| \ll |\alpha_0|, \quad \beta \approx \beta_0 + \beta_1(t), \quad |\beta_1| \ll |\beta_0|$$

$$\dot{\alpha}_1 = -\frac{4\lambda\alpha_0^2}{k^2} \sin^3(\omega t + \beta_0) \cos(\omega t + \beta_0) \quad \Rightarrow \quad \alpha_1(t) = -\frac{\lambda\alpha_0^2}{k^2\omega} \sin^4(\omega t + \beta_0)$$

$$\dot{\beta}_1 = \frac{2\lambda\alpha_0}{k^2} \sin^4(\omega t + \beta_0)$$

$$\Rightarrow \beta_1(t) = \frac{3\lambda\alpha_0}{4k^2\omega^2} (\omega t + \beta_0) + \frac{\lambda\alpha_0}{16k^2\omega^2} \frac{\sin(4(\omega t + \beta_0)) - 8\sin(2(\omega t + \beta_0))}{32\omega}$$

Full solution: $\alpha_1(t)$ affects amplitude of oscillations

$\beta_1(t)$ affects **phase** and **frequency** of oscillation:

$$\sin(\omega t + \beta) = \sin(\omega t + \beta_0 + \beta_1) =$$

$$\sin \left[\left(\omega + \frac{3\lambda\alpha_0}{4k^2\omega} \right) t + \beta_0 \left(1 + \frac{3\lambda\alpha_0}{4k^2\omega^2} \right) + \frac{\lambda\alpha_0}{16k^2\omega^2} \frac{\sin(4(\omega t + \beta_0)) - 8\sin(2(\omega t + \beta_0))}{32\omega} \right]$$

Example 2

Frequency change:

$$\omega \rightarrow \left(\omega + \frac{3\lambda\alpha_0}{4k^2\omega} \right) \quad (1)$$

Compare with our earlier result (“ordinary” theory of oscillations, anharmonic terms)

$$\omega \rightarrow \omega + \frac{3\lambda |\mathbb{A}_0|^2}{2\omega_0 m} \quad (2)$$

Can see that results coincide, if we recall that the amplitude of oscillations is $2|\mathbb{A}_0|$

$$x_0(t) = \operatorname{Re} (\mathbb{A}_0 e^{i\omega t} + \mathbb{A}_0^* e^{-i\omega t}) = 2|\mathbb{A}_0| \cos(\omega t + \arg \mathbb{A}_0)$$

and is related to energy (parameter $\alpha \equiv E$) as

$$2\mathbb{A}_0 = \sqrt{\frac{2\alpha}{k}}, \quad \alpha = 2k |\mathbb{A}_0|^2$$

Example 3

A particle of mass m scatters in the central potential

$$U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^n}, \quad \alpha > 0$$

Develop a canonical perturbation theory and find trajectory assuming that the last term $\sim \mathcal{O}(1/r^n)$ is a small correction.

Example 3

A particle of mass m scatters in the central potential

$$U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^n}, \quad \alpha > 0$$

Develop a canonical perturbation theory and find trajectory assuming that the last term $\sim \mathcal{O}(1/r^n)$ is a small correction.

Before we start perturbative analysis, try to analyze yourself:

What integrals of motion do we have in this field ?

Are Kepler's laws valid in this potential?

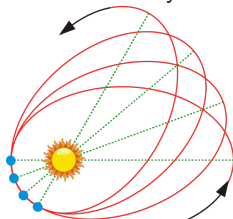
How the trajectories should look like?



Example 3

We'll consider perturbation $\Delta H = \frac{\kappa}{r^n}$

- ▶ for $\Delta H = 0$ the bound state solution ($E < 0$) corresponds to motion over elliptic orbit
- expect due to ΔH the parameters of orbit will change slowly
 - ...the motion will remain in plane because H is still central potential
 - ... yet the orbit will slowly rotate (precession of periapsis)



– Now we'll recall HJ formalism for Coulomb field

HJ formalism for central field (previous lectures)

HJ for motion in central field

$$U(r, \theta, \phi) = a(r)$$

- Energy is conserving:

$$\Rightarrow S_0 = f(r, \theta, \phi) - E t, \quad E \equiv \alpha_1$$

- $f(r, \theta, \phi) = f_3(\phi) + f_2(\theta) + f_1(r)$

$$\Rightarrow f_3(\phi) = \sqrt{2m\alpha_3}(\phi - \phi_0)$$

$$\Rightarrow f_2(\theta) = \sqrt{2m} \int d\theta \sqrt{\frac{\alpha_3}{\sin^2 \theta} - \alpha_2} + \theta_0$$

—result for $f_2(\theta)$ is complicated elementary function

$$\Rightarrow f_1(r) = \int dr \sqrt{2m \left(E - a(r) + \frac{\alpha_2}{r^2} \right)}$$

—Conjugate “coordinates” (variables β_i): t_0, θ_0, ϕ_0

HJ formalism for central field (previous lectures)

- "Momenta" constants α_i : ($\alpha_1 \equiv E$, α_2 , α_3)

$$\frac{\partial f_1(r, E)}{\partial E} - t = \text{const} = -t_0,$$

$$\Rightarrow 2m \int \frac{dr}{\sqrt{2m(E - a(r) - \frac{\alpha_2}{r^2})}} = t - t_0$$

$\Rightarrow r = r(t)$ - dependence. Coincides with what we had if $\alpha_2 \equiv M^2/2m$

–Now we'll assume that $\alpha_3 = 0$ (You may find full consideration without $\alpha_3 = 0$ assumption in Section 10.8 in Goldstein)

$$\Rightarrow f_2(\theta) = \sqrt{2m\alpha_2}i(\theta - \theta_0)$$

$\Rightarrow \partial S(r, E)/\partial \alpha_2$ will yield familiar

$$\theta - \theta_0 = \int^r dr \frac{M}{m r^2 \sqrt{\frac{2}{m}(E - U_{\text{eff}}(r))}} \quad (3)$$

in agreement with our earlier result

Example 3 (continued)

- ▶ Since we have central potential, trajectory always remains in plane, ΔH will cause precession of periastr (θ_0 will depend slowly on time t)
- ▶ Conjugate “momentum” (variable α_i) after we make canonical transformation with S_0 :

$$\frac{\partial S_0}{\partial \theta_0} \equiv \frac{\partial f_2}{\partial \theta_0} = \sqrt{2m\alpha_2} = M$$

- angular momentum of the particle
- Canonical perturbation theory for $\dot{\theta}_0$:

$$\dot{\theta}_0 \equiv \left. \frac{\partial \Delta H}{\partial M} \right|_{r=r_0(t, E, M)}$$

- dependence on M appears because trajectory r_0 depends on it
- In general $\dot{\theta}_0$ depends on time, not homogeneous rotation with const. angular velocity. Let's evaluate explicitly *average* (per period) change of $\dot{\theta}$

$$\overline{\dot{\theta}_0} = \frac{\overline{\partial \Delta H}}{\partial M} = \frac{\partial \overline{\Delta H}}{\partial M}, \quad \overline{\Delta H} \equiv \frac{1}{T} \int_0^T dt \Delta H = \frac{\kappa}{T} \int_0^T \frac{dt}{r^n(t)}$$

For the first order correction should use $r(t)$ found in $\Delta H = 0$ limit

Example 3 (continued)

► 2nd law of Kepler: $M dt = mr^2 d\theta$

$$\overline{\Delta H} \equiv \frac{1}{T} \int_0^T dt \Delta H = \frac{\kappa}{T} \int_0^T \frac{dt}{r^n(t)} = \frac{m}{MT} \int_0^{2\pi} \frac{d\theta}{r^{n-2}(\theta)}$$

For the first order correction should use $r(\theta)$ found in $\Delta H = 0$ limit, i.e. solution of Kepler's problem:

$$\frac{p}{r(\theta)} = 1 + e \cos(\theta - \theta_0)$$

$$p = \frac{M^2}{m\alpha}, \quad e = \sqrt{1 + \frac{2EM^2}{m\alpha^2}}$$

$$\overline{\Delta H} \equiv \frac{m\kappa}{T} \frac{(m\alpha)^{n-2}}{M^{2n-3}} \int_0^{2\pi} d\theta (1 + e \cos \theta)^{n-2}$$

Example 3 (continued)

$$\overline{\Delta H} \equiv \frac{m\kappa}{T} \frac{(m\alpha)^{n-2}}{M^{2n-3}} \int_0^{2\pi} d\theta (1 + e \cos \theta)^{n-2}$$

Evaluation of precession:

–Case $n = 2$:

$$\overline{\Delta H} = 2\pi \frac{\kappa m}{M T} \quad \Rightarrow \quad \dot{\theta}_0 = \frac{\partial \overline{\Delta H}}{\partial M} = -\frac{2\pi \kappa m}{M^2 T}$$

–Case $n = 3$ (General Relativity correction):

$$\overline{\Delta H} \equiv 2\pi \frac{m^2 \alpha \kappa}{M^3 T} \quad \Rightarrow \quad \dot{\theta}_0 = \frac{\partial \overline{\Delta H}}{\partial M} = -\frac{6\pi m^2 \alpha \kappa}{M^4 T}$$

–precession of orbits of planets,

–largest for Mercury (0.43"/year), one of the “classical” tests which helped to confirm GR

Comparison with data, should sum up contributions of all perturbations:

Relativistic correction, perturbation due to other planets, etc (not trivial).

►General case (arbitrary n): lengthy expression with hypergeometric functions ${}_2F_1(\dots)$

Example 4: Apsides precession

A satellite moves in the gravitational field of the Earth

$$U(r) = -\frac{\alpha}{r} + \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

where θ is the angle between the symmetry axis of the Earth (oblate spheroid) and the radius-vector pointing towards the satellite. Assuming that the last term (quadrupole interaction) is a small parameter, evaluate the precession of the orbits of the satellites

Before we start perturbative analysis, try to analyze yourself:

What integrals of motion do we have in this field ?

Are Kepler's laws valid in this potential?

How the trajectories should look like?



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► Potential in general not analytically solvable:

Euler-Lagrange: coupled nonlinear equations.

Hamilton-Jacobi: separation of variables doesn't work

Caution: *We have seen that in absence of perturbation in the Coulomb field the orbits are plane ellipsoids. However, these planes do not have to coincide with plane $\theta = \pi/2$ now, may be inclined. We need to learn how to relate different angles in reference frames*

Example 4: Apsides precession

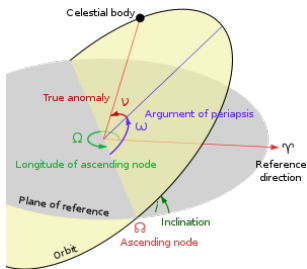
► Split H as $H = H_0 + \Delta H$, perturbation:

$$\Delta H = \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

challenge for perturbation theory: explicit dependence on angle θ

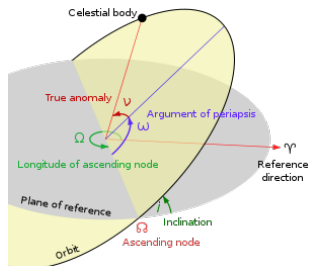
⇒ periapsis will precess, but result will depend on orientation

⇒ plane of motion will also precess ($\mathbf{M} \neq \text{const}$)



sometimes this is called apsidal precession (=precession of periapsis or apoapsis)

Some geometrical definitions

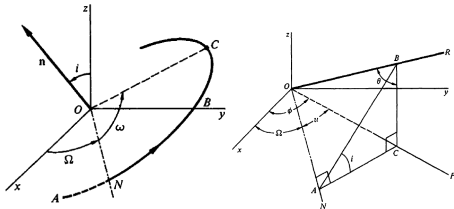


► Let's define Delaunay's orbital elements (angles) as shown in the Figure:

- ω is the position of periaapsis in the plane of motion
- ψ is the polar angle of particle *in plane of motion*
- i is the *inclination* - angle between plane of motion and plane $\perp$ to axis of spheroid
- Ω is the longitude of ascending node
- "True anomaly", "epoch": angle $\psi + \omega$

Some geometrical definitions (II)

Relation between angles (from *Goldstein*):



$$OB = 1, BC = \cos \theta, AB = \sin (\psi + \omega)$$

$$\Rightarrow BC = \sin (\psi + \omega) \sin i$$

$$\cos \theta = \sin i \sin (\psi + \omega)$$

$$\Rightarrow 3 \cos^2 \theta - 1 = \frac{1 - 3 \cos^2 i}{2} - \frac{3}{2} \sin^2 i \cos 2(\psi + \omega)$$

Example 4: Apisides precession

Perturbation theory:

$$\Delta H = \frac{\kappa}{r^3} (3 \cos^2 \theta - 1)$$

$$\cos \theta = \sin i \sin (\theta + \theta_0)$$

$$\Rightarrow 3 \cos^2 \theta - 1 = \frac{1 - 3 \cos^2 i}{2} - \frac{3}{2} \sin^2 i \cos 2(\psi + \omega)$$

● Period-averaged correction:

$$\begin{aligned} \overline{\Delta H} &\equiv \frac{m^2 \kappa}{M^3 T} \int_0^{2\pi} d\psi (1 + e \cos \psi) (3 \cos^2 \theta - 1) = \\ &\equiv \frac{m^2 \kappa}{M^3 T} \int_0^{2\pi} d\psi (1 + e \cos \psi) \left(\frac{1 - 3 \cos^2 i}{2} - \frac{3}{2} \sin^2 i \cos 2(\psi + \omega) \right) \end{aligned}$$

Example 4: Apssides precession

$$\int_0^{2\pi} d\psi (1 + e \cos \psi) \cos 2(\psi + \omega) = 0$$

$$\overline{\Delta H} \equiv 2\pi \frac{m^2 \kappa}{M^3 T} \frac{1 - 3 \cos^2 i}{2}$$

result for rotation of periastr:

$$\Rightarrow \dot{\omega} = \frac{\partial \overline{\Delta H}}{\partial M} = -\frac{6\pi m^2 \alpha \kappa}{M^4 T} \frac{1 - 3 \cos^2 i}{2}$$

► Angle i determines orientation of motion plane (ϕ_0 in our previous HJ formulas)
... it is canonically conjugate to parameter α_3 (we used $\alpha_3 = \text{const} = 0$ for fixed motion in plane, but now can relax this condition).

– α_3 is related to M_z as $M_z \equiv \sqrt{2 m \alpha_3} \equiv M \cos i$

$$\Rightarrow \left(\frac{di}{dt} \right) = \frac{\partial \overline{\Delta H}}{\partial M_z} = \frac{\partial \overline{\Delta H}}{M \partial \cos i} = -\frac{6\pi m^2 \kappa}{M^4 T} \cos i$$

– Mercury precession: the quadrupole moment of the Sun is small, so precession due to GR is numerically larger

Summary: Perturbation theory

- 1 Assume $H = H_0 + \Delta H$, and for H_0 have analytic solutions
 - 1 Use Hamilton-Jacobi-formalism, $S_0(q_1, \dots, q_N, \alpha_1, \dots, \alpha_N)$ as generating function
 - 2 Variables $\alpha_1, \dots, \alpha_N, \beta_1, \dots, \beta_N$ are constants in system with hamiltonian H_0 , but NOT constants in theory with hamiltonian H
 - 3 Substitute trajectories $\{p_0, q_0\}$ of H_0 into $\Delta H = H - H_0$
- 2 Solve canonical equations for (α, β) :

$$\dot{\alpha}_a = -\frac{\partial \Delta H}{\partial \beta_a}, \quad \dot{\beta}_a = \frac{\partial \Delta H}{\partial \alpha_a},$$

- 1 Usually we can't solve (2) exactly
- 2 ... but we know that ΔH is small, so can assume that $\alpha(t), \beta(t)$ *change slowly*:

$$\alpha \approx \alpha_0 + \alpha_1(t) + \alpha_2(t) + \dots, \quad |\alpha_2| \ll |\alpha_1| \ll |\alpha_0|$$

$$\beta \approx \beta_0 + \beta_1(t) + \beta_2(t) + \dots, \quad |\beta_2| \ll |\beta_1| \ll |\beta_0|$$

α_0, β_0 are constants, and $\alpha_1, \dots, \beta_1, \dots$ can be found using perturbative expansion over ΔH .